

ASSIGNMENT 02**Due date: Monday, 30 June 2025**

Total Marks: 20

ONLY FOR YEAR MODULE

This assignment covers **chapters 1 & 2 of the prescribed book and the study guide**, it is specifically based on Sections 1.1 - 1.7 of Chapter 1 and Chapter 2, **Note that this assignment will close on 27 June 2026 at 11:00 PM**

Question 1: 6 Marks

Let A be an $n \times n$ matrix such that $A^3 - 3A^2 + 3A - I = 0$, where I is the $n \times n$ identity matrix.

- Prove that A is invertible.
- Find an expression for A^{-1} in terms of A and I .
- Find an expression for A^4 in terms of A and I .

Question 2: 6 Marks

Let A be a 3×3 matrix. Let $p(x) = \det(A - xI)$ be the characteristic polynomial of A . If $p(x) = -x^3 + 2x^2 - 5x + 8$,

- Find the determinant of A .
- Use the Cayley-Hamilton theorem to find the inverse of A .
- Find the trace of A .

Cayley-Hamilton theorem Any matrix A satisfies its own characteristic equation.

Question 3: 8 Marks

Let A and B be $n \times n$ matrices. Prove that if $I - AB$ is invertible, then $I - BA$ is also invertible and

$$(I - BA)^{-1} = I + B(I - AB)^{-1}A$$